Six winter plans

How Animals Get Ready for Winter

Name _____ Date _____



Big and Little walked to the edge of the woods. The days were getting short. A chickadee tucked a seed into a crack in the bark, then flew to another tree. Big said chickadees hide food in many spots in fall. They use a part of the brain called the hippocampus to find those spots again later.



Under the leaves, a wood frog freezes. Ice forms in its body, but outside its cells. Its liver pours out glucose, a kind of sugar, that helps protect the cells. A woolly bear caterpillar makes glycerol, an antifreeze-like protectant. In a lab test, caterpillars cooled slowly for weeks moved again after a week frozen at about -4 degrees Fahrenheit.



In winter, a chipmunk's body gets much colder on purpose. This is called torpor, and it saves energy. For most small hibernators, hibernation is many long torpor stretches with short warm-ups between them. Many mammal hibernators burn brown fat, a special fat that makes heat, to warm back up. A bear is different. It cools only a few degrees.

hippocampus	a brain area chickadees use to find their hidden food
glycerol	an antifreeze-like protectant a woolly bear makes
glucose	a kind of sugar that a wood frog's liver sends out to protect its cells
torpor	a time when an animal lets its body get cold and slow to save energy
hibernation	in most small mammal hibernators, long torpor stretches with short warm-ups
brown fat	a special fat in some mammals that burns fuel to make heat
cluster	a tight ball of honeybees that keeps the middle warm
subnivean zone	the space between the ground and the snow above it

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Honeybees do not hibernate. In winter they pack together into one ball called a cluster. The bees on the outside make a living blanket. The bees in the middle shiver their flight muscles without moving their wings. That shivering makes heat. When there are no baby bees, the middle stays warmer than about 64 degrees Fahrenheit.



Little found a tiny hole in the snow. Big said it was an air hole. Under the snow, voles and mice dig tunnels along the ground. This space is called the subnivean zone. About six inches of snow works like a blanket, so it stays near 32 degrees Fahrenheit there, even when the air above is much colder. The air holes let carbon dioxide out.



In snowy places, most snowshoe hares grow a white coat in fall. In one study, the change started on about the same dates every year, snow or no snow. Scientists think the length of the day is the main trigger. The change takes about 40 days. So when snow comes late, a white hare can end up sitting on brown leaves.

1. How does a wood frog stay alive when ice forms in its body?
2. How are a chipmunk's plan and a honeybee's plan for staying alive different?
3. Why is the space under the snow warmer than the air, and why can a white hare end up on brown leaves?

Your turn

Pick one animal. Write one sentence that tells its winter plan.

Example: Honeybees huddle in a ball, so the bees in the middle stay warm.